Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 11)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change .....

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Ozone depletion .....

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Tourism .....

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Future mineral exploitation .....

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0 1 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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0 2 . 1 Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition

Soil maintenance

Interspecies relationships

0 2 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

0 2 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

10

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0 3 . 1 Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0 3 . 2 Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

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0 4 Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO_2, NOx

0 4 . 1 Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy.

[4 marks]

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0 4 . 2 Give **three** other natural processes driven by solar energy that human society depends on.

[3 marks]

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0 4 . 3 Explain how the principle of low-temperature natural processes is being applied in human systems.

[3 marks]

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0 5 Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

0 5 . 1 Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

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0 5 . 2 Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

0 5 . 3 Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

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0 6 . 1 Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

0 6 . 2 Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

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0 6 . 3 Explain how auditory monitoring can be used to study bats.

[2 marks]

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0 6 . 4 Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

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0 7 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 7 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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0 7 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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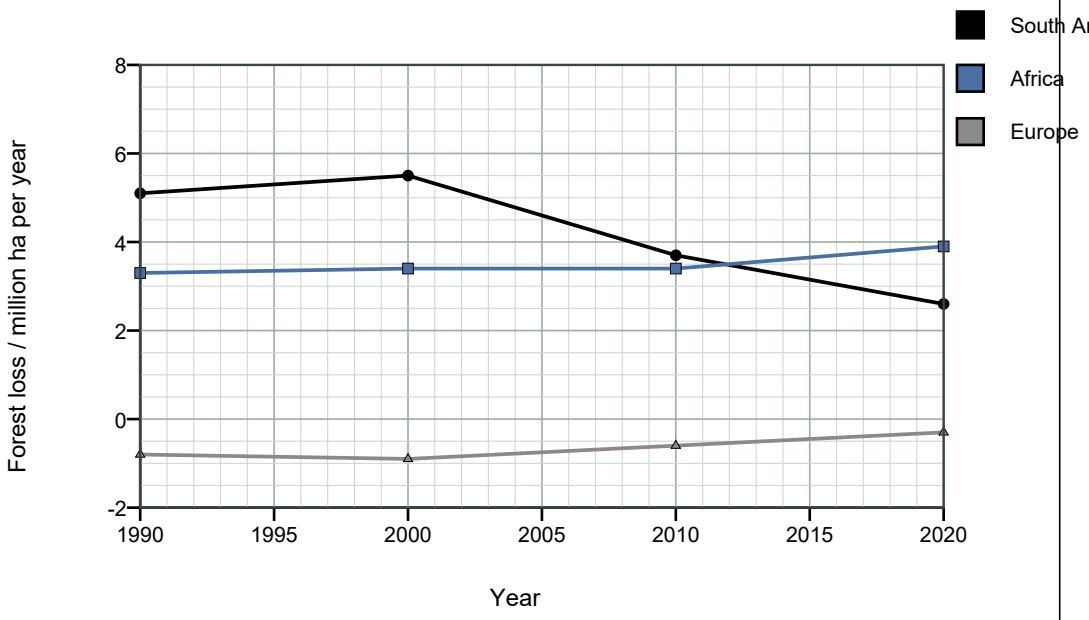
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0 8 Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 8 . 1 Describe the trends shown in Figure 1.

[3 marks]

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0 8 . 2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

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0 8 . 3 Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

0 8 . 4 Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

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0 9 . 1 Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

0 9 . 2 Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

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0 9 . 3 Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

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1 0 . 1 Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

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Extra space

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1 0 . 2 State what is meant by **permaculture**.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

OR

11 . 2 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS